

Erdős Problem #788

EP-788

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $f(n)$ be maximal such that if $B \subset (2n, 4n) \cap \mathbb{N}$ there exists some $C \subset (n, 2n) \cap \mathbb{N}$ such that $c_1 + c_2 \notin B$ for all $c_1, c_2 \in C$ and $|C| + |B| \geq f(n)$.

Estimate $f(n)$. In particular is it true that $f(n) \leq n^{1/2+o(1)}$?